EDC BLOCK-003 Gravity Sector: Closure Summary under Minimal Calibration

EDC Collaboration

Program note / consolidation; no new results

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Abstract

This note consolidates derivations v13–v17 into a single reader-friendly summary of the BLOCK-003 gravity sector closure. We present the canonical derivation chain, numerical results, robustness checks, and an explicit "reader contract" specifying what is derived [D], what is identified [I], and what remains baseline-calibrated [BL]. **No new results are claimed**; this is a consolidation document.

1 Scope and Reader Contract

This document establishes a clear epistemic contract between EDC and the reader regarding the gravity sector:

- **What is derived [D]:** The 5D-to-4D reduction formula $M_{\rm Pl}^2 = M_5^3 \mathcal{I}$ follows from standard Kaluza-Klein dimensional reduction applied to the EDC 5D Einstein-Hilbert action. This is mathematical physics, not phenomenological fitting.
- **What is identified [I]:** The compactification scale R_ξ is *identified* with an electroweak length scale via $R_\xi = \hbar c / M_Z^{\rm obs}$. This identification is physically motivated but not derived from EDC axioms.
- **What is baseline-calibrated [BL]:** Two measured quantities serve as baseline inputs: the 4D Planck mass $M_{\rm Pl}^{\rm obs} = 1.221 \times 10^{19}$ GeV and the Z boson mass $M_Z^{\rm obs} = 91.1876$ GeV. These are taken from experiment.
- **What remains NO-GO:** A purely internal derivation of R_ξ from EDC geometry (Track A in v16) is not possible with current axioms. Full predictive closure would require deriving R_ξ without EW input.

2 Canonical Derivation Chain

The gravity sector closure proceeds through five key equations:

2.1 Step 1: Normalization Extractor (v13)

From weak-field 5D→4D matching, the 4D Planck mass is determined by the 5D Planck mass M_5 and a dimensionless integral $\mathcal{I}$:

$$\boxed{M_{\rm Pl}^2 = M_5^3 \, \mathcal{I}} \tag{1}$$

where $\mathcal{I} = \int d\xi \, e^{4A(\xi)} |\psi_0(\xi)|^2$ encodes the warp profile $A(\xi)$ and graviton zero-mode $\psi_0(\xi)$. (tag: [D])

2.2 Step 2: Newton's Constant (v13)

The effective 4D Newton constant follows:

$$G_N = \frac{1}{8\pi M_5^3 \mathcal{I}} \tag{2}$$

This reproduces the standard $G_N = 1/(8\pi M_{\rm Pl}^2)$ relation. (tag: [D])

2.3 Step 3: Integral Evaluation — Model A (v14)

In the preferred compact model (Model A: flat extra dimension with $A(\xi) = 0$, $\psi_0 = \text{const}$), the integral evaluates to:

$$\mathcal{I} = R_\xi \tag{3}$$

where R_ξ is the compactification radius. (tag: [D] conditional on model choice)

2.4 Step 4: EW-Scale Identification (v16–v17)

The compactification scale is identified with an electroweak length:

$$\boxed{R_\xi = \frac{\hbar c}{M_Z^{\rm obs}}} \tag{4}$$

with canonical choice $M_Z^{\rm obs} = 91.1876 \pm 0.0021$ GeV. (tag: [I]+[BL])

2.5 Step 5: 5D Planck Mass (v15–v16)

Combining Eqs. (1)–(4):

$$\boxed{M_5 = \left(\frac{M_{\rm Pl}^2}{R_\xi}\right)^{1/3} = M_{\rm Pl}^{2/3}\, M_Z^{1/3}} \tag{5}$$

This is the calibrated closure formula. (tag: [D] given [I]+[BL] inputs)

3 Numerical Closure

3.1 Baseline Inputs

Table 1: Baseline inputs for gravity sector closure.

Quantity	Symbol	Value	Tag
4D Planck mass	$M_{\rm Pl}^{\rm obs}$	1.221×10^{19} GeV	[BL]
Z boson mass	$M_Z^{\rm obs}$	91.1876 ± 0.0021 GeV	[BL]

3.2 Derived Outputs

Table 2: Derived quantities from calibrated closure.

Quantity	Symbol	Value	Tag
Compactification scale	R_ξ	2.165×10^{-18} m	[I]+[BL]
5D Planck mass	M_5	2.41×10^{13} GeV	[D]

3.3 Error Budget

From error propagation (v15–v16):

$$\frac{\delta M_5}{M_5} = \sqrt{\left(\frac{2}{3}\frac{\delta M_{\rm Pl}}{M_{\rm Pl}}\right)^2 + \left(\frac{1}{3}\frac{\delta M_Z}{M_Z}\right)^2} \approx 1.1 \times 10^{-5} \tag{6}$$

Closure statement: The gravity sector is CLOSED (calibrated) under minimal baseline inputs: $M_{\rm Pl}^{\rm obs}$ [BL] and $M_Z^{\rm obs}$ [BL]. The 5D Planck mass is $M_5 = 2.41 \times 10^{13}$ GeV $\pm 0.001\%$.

4 Robustness Check (v17)

The identification $R_\xi = \hbar c/M_*$ was tested with alternative EW scales:

Table 3: Calibration family robustness (from v17).

M_*	Value (GeV)	M_5 (GeV)	$\Delta \log_{10} M_5$	Status
M_Z (canonical)	91.19	2.41×10^{13}	0 (ref)	[BL]
M_W	80.38	2.31×10^{13}	-0.018	[Dc]
$v_{\rm EW}$	246.2	3.35×10^{13}	$+0.143$	[BL]

Verdict: All choices yield M_5 in the range (2.3–3.4) $\times 10^{13}$ GeV, with maximum shift $|\Delta \log_{10} M_5| = 0.143$ (factor ~ 1.4). The calibrated closure is **order-of-magnitude stable** under reasonable alternative EW-scale identifications.

5 What Remains Open

5.1 Track A: Internal Derivation (NO-GO)

Derivation v16 attempted to derive R_ξ from EDC axioms alone (Track A). **Result: NO-GO.** All candidate relations in the repository either:

1. Use R_ξ as input, not output
2. Are postulates [P], not derivations
3. Relocate the unknown to another undetermined quantity

5.2 What Would Constitute Full Closure

Fully predictive closure would require one of:

- Deriving M_Z (or R_ξ) from membrane dynamics
- Finding an independent EDC observable that constrains R_ξ
- Deriving R_ξ directly from the EDC action without EW input

None of these routes are currently available. The gravity sector remains **calibrated** (not fully predicted).

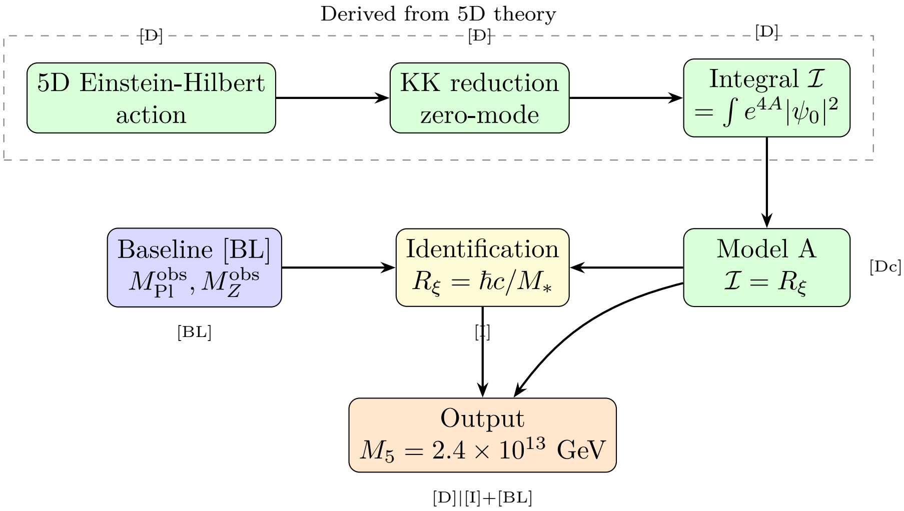

Figure 1: Gravity sector derivation pipeline. Green boxes: derived [D]. Yellow: identified [I]. Blue: baseline inputs [BL]. Orange: output. **Schematic only; not to scale; no new results; consolidation of v13–v17.**

Table 4: Complete epistemic ledger for BLOCK-003 gravity closure.

Element	Formula/Value	Tag	Source
Normalization extractor	$M_{\mathrm{Pl}}^2 = M_5^3 \mathcal{I}$	[D]	v13
Newton constant	$G_N = 1/(8\pi M_5^3 \mathcal{I})$	[D]	v13
Integral (Model A)	$\mathcal{I} = R_\xi$	[Dc]	v14
EW identification	$R_\xi = \hbar c/M_Z$	[I]	v16
Z mass input	$M_Z = 91.1876$ GeV	[BL]	PDG
Planck mass input	$M_{\mathrm{Pl}} = 1.221 \times 10^{19}$ GeV	[BL]	CODATA
Closure formula	$M_5 = M_{\mathrm{Pl}}^{2/3} R_\xi^{-1/3}$	[D]	v15
Numerical result	$M_5 = 2.41 \times 10^{13}$ GeV	[D]||[BL]	v16
Robustness	$\Delta \log_{10} M_5 < 0.15$	[D]	v17
Internal derivation	R_ξ from axioms	**NO-GO**	v16

6 Derivation Pipeline

7 Epistemic Ledger

Summary

BLOCK-003 Gravity Sector Status:

CLOSED (calibrated) under minimal baseline inputs ($M_{\mathrm{Pl}}^{\mathrm{obs}}$, M_Z^{obs}).

NOT closed as pure prediction: R_ξ identification is [I]+[BL], not [D].

Result: $M_5 = 2.41 \times 10^{13}$ GeV (GUT scale) $\pm 0.001\%$.

Robustness: Verified across EW-scale family (v17).

Open item: Internal derivation of R_ξ remains NO-GO.

This document consolidates derivations v13–v17; no new results are claimed.